Digital Ecosystems & Governance in Organizations Group Project: · Credit Application Governance Analysis – NovaCred

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Introduction & Executive Summary

GOAL: Audit the dataset to identify data quality issues, potential algorithmic bias, and governance risks in automated credit decisions.

PROJECT: The project is structured into three parts: data quality assessment, bias analysis in loan approvals, and privacy and governance evaluation of the credit decision system.

DATASET: The raw dataset contains over 500 credit applications with applicant information, financial attributes, spending behavior, and loan decision outcomes.

Data Quality Findings

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Total issues catalogued: 20
By dimension:
dimension
Completeness     7
Validity         6
Consistency      4
Accuracy         3
```

	dimension	column	issue
0	Completeness	decision_notes	99.6% missing
1	Completeness	processing_timestamp	87.7% missing
2	Completeness	applicant_info_email	1.4% missing
3	Completeness	financials_annual_income	1.0% missing
4	Completeness	applicant_info_ssn	1.0% missing
5	Completeness	applicant_info_dob	1.0% missing
6	Completeness	applicant_info_gender	0.6% missing
7	Consistency	applicant_info_gender	M/F vs Male/Female mixed encodings
8	Consistency	applicant_info_id	Duplicate application IDs
9	Consistency	applicant_info_ssn	Duplicate SSNs across records
10	Consistency	decision_notes	DUPLICATE_ENTRY_ERROR marker records
11	Validity	financials_credit_history_months	Negative values (e.g. -10, -3) — impossible duration
12	Validity	financials_debt_to_income	Values > 1.0 (e.g. 1.85) — exceeds valid ratio ceiling
13	Validity	financials_savings_balance	Negative value (-5000) — impossible
14	Validity	financials_annual_income	Zero value — economically implausible for credit applicant
15	Validity	applicant_info_email	Invalid format (missing @, no domain, etc.)
16	Validity	applicant_info_dob	Age outside 18–100 range or unparseable date string
17	Accuracy	processing_timestamp	Future-dated timestamps — system clock / pre-loaded records
18	Accuracy	financials_annual_income	String-encoded in source JSON — numeric field stored as wrong type
19	Accuracy	applicant_info_zip_code	Read as float64 (e.g. 10036.0) — must be zero-padded 5-char string

CHECK NEXT PAGE FOR FINDINGS

What we fixed

7
COMPLETENESS

- decision_notes — 99.6% null → dropped
- processing_timestamp — 87.7% null → flagged
- email, SSN, DOB, income — ~1% missing
- gender — 0.6% missing

4
CONSISTENCY

- Gender: M/F → Male/Female standardised
- Duplicate application IDs removed
- Duplicate SSNs deduplicated
- DUPLICATE_ENTRY_ERROR notes removed

6
VALIDITY

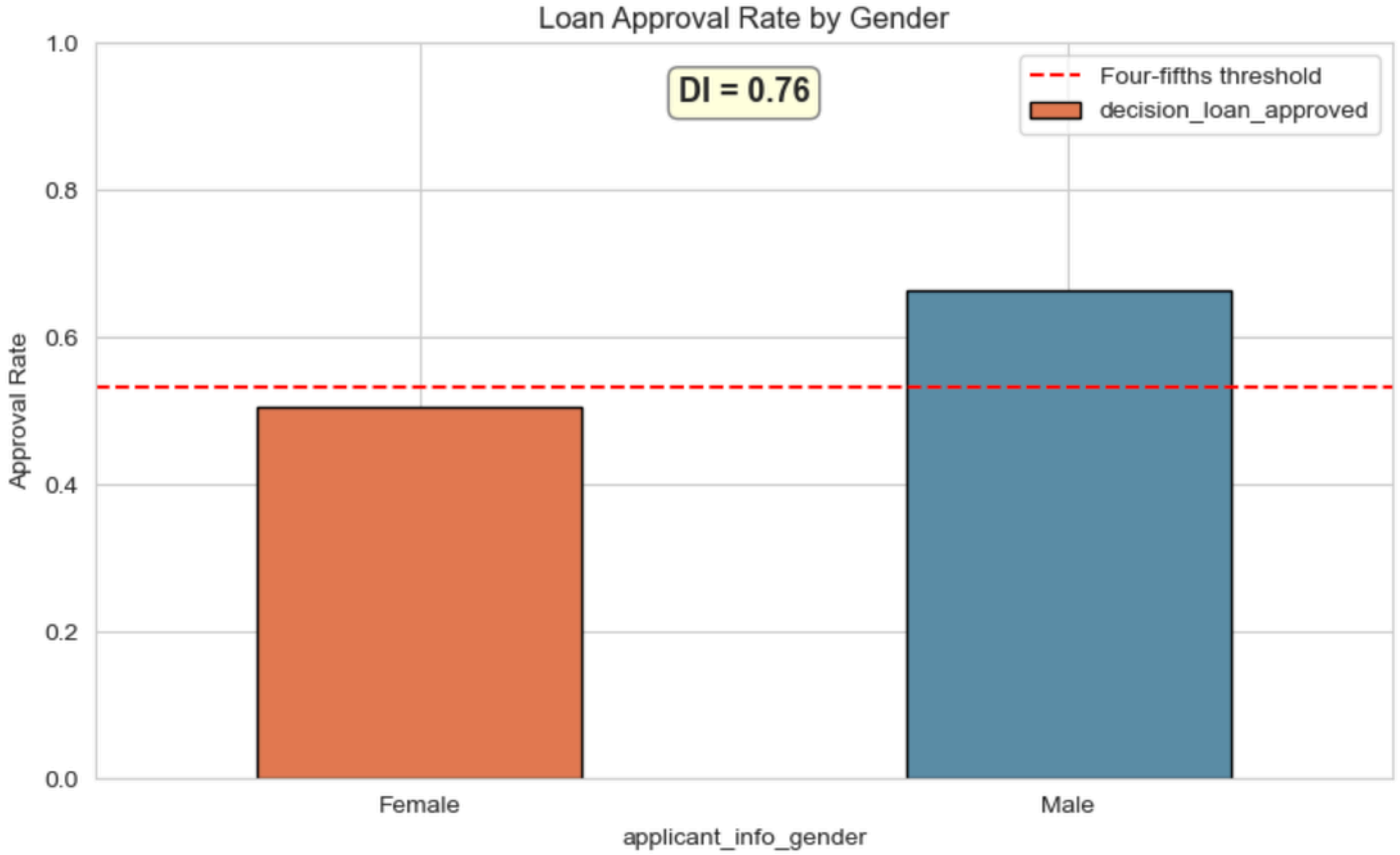
- Negative credit history months → null
- DTI > 1.0 (e.g. 1.85) → null
- Negative savings balance → null
- Zero income flagged & nulled
- Invalid emails nulled
- Age outside 18–100 range → null

3
ACCURACY

- Future timestamps → flagged
- Income: string→numeric explicit cast
- Zip code: float→zero-padded string

Bias Analysis

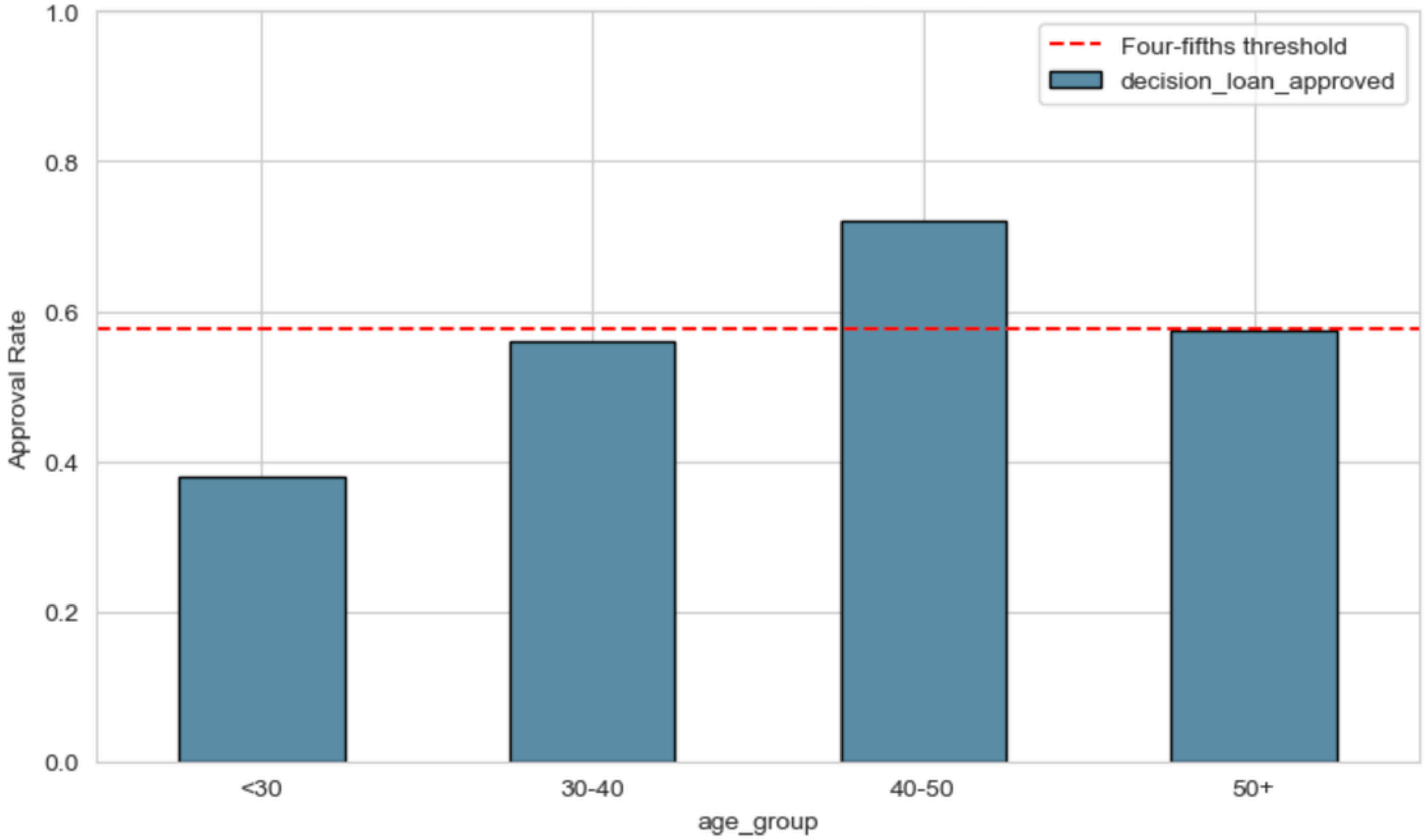
Gender



Approval rates by gender

- DI ratio: 0.76
- Chi-squared test significant
- Female applicants are significantly less likely to be approved, indicating potential algorithmic bias

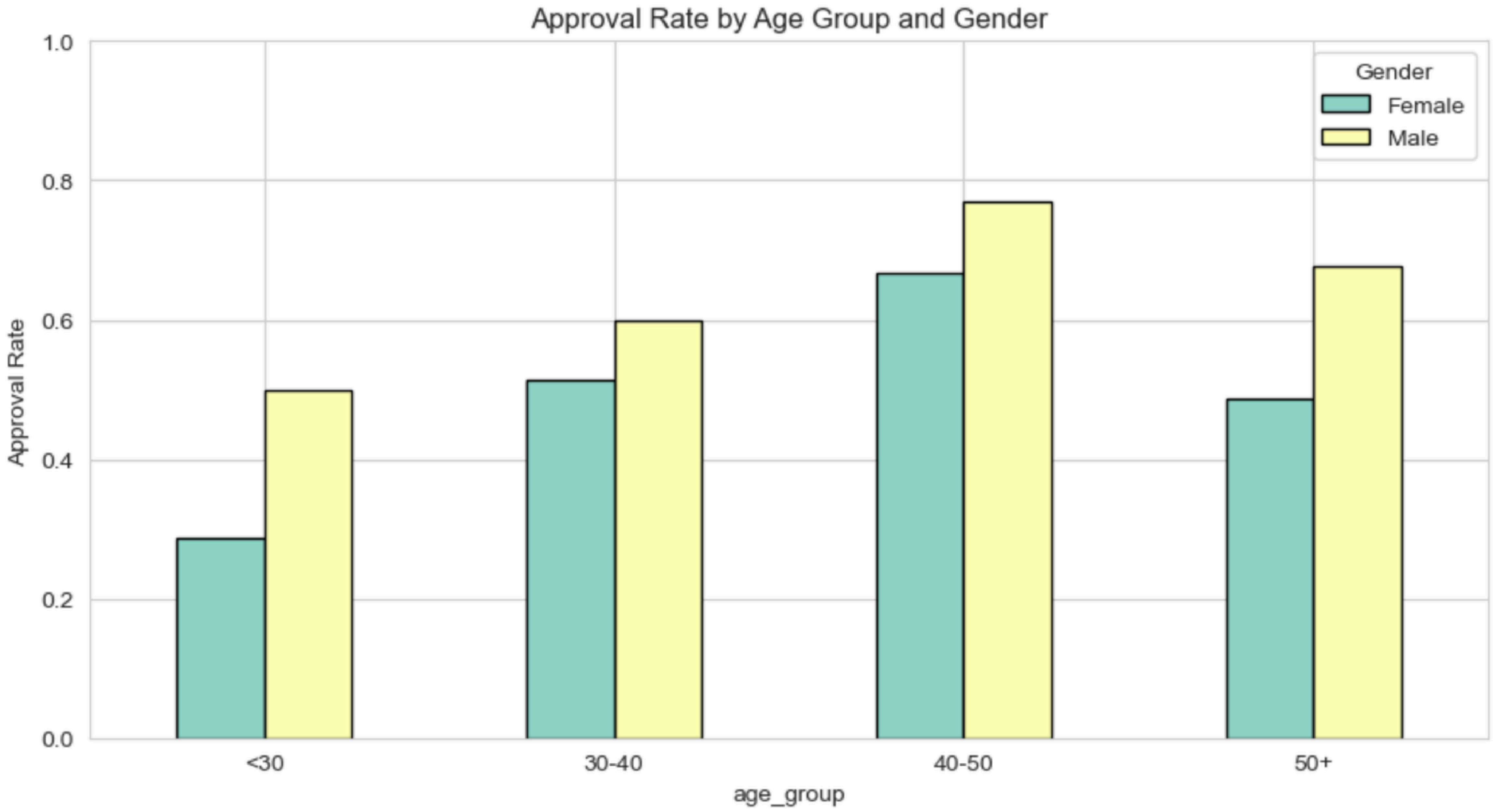
Age



Approval rates by age

- DI ratio: 0.53
- Chi-squared test significant
- The algorithm strongly favours 40-50 age group, while young applicants are disadvantaged

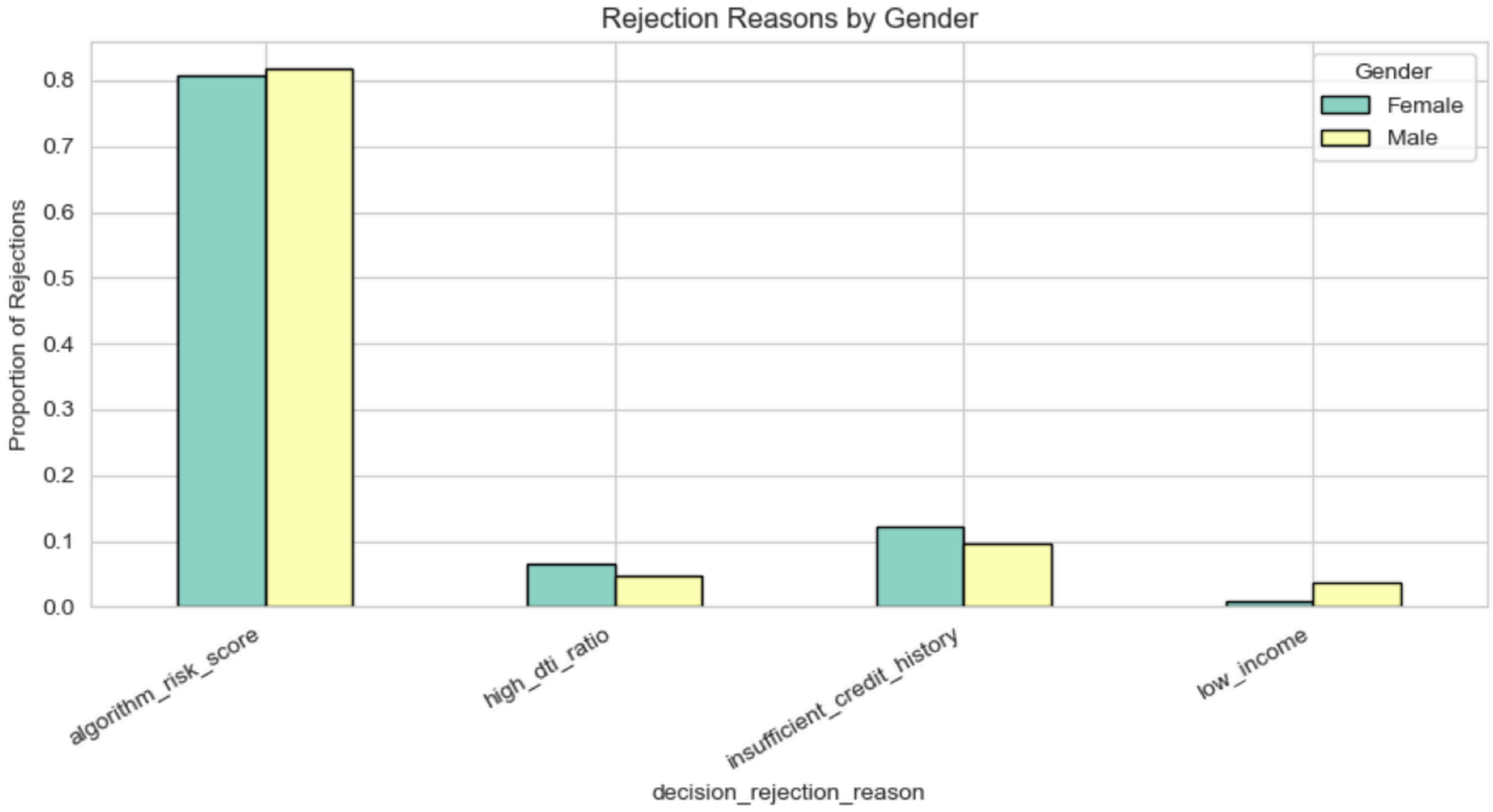
Bias Analysis



Bias becomes much stronger when age and gender interact:

- Young females (<30) have the lowest approval rate (~28.6%).
- Males aged 40–50 have the highest approval (~76.9%).
- Disparate Impact ratio = 0.37 between these groups.

Bias Analysis



- ~81% of rejections are due to the algorithm risk score
- ~39.8% of females are rejected by the algorithm compared to ~27.4% of males
- under-30 group faces a 47.3% algorithm rejection rate versus only 23.0% for the 40–50 group
- Bias is concentrated in the algorithmic scoring model itself.

Bias Analysis

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BIAS ANALYSIS SUMMARY
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	Test	Value	Threshold	Result
0	Gender DI ratio	0.7605	< 0.80	FAIL
1	Gender chi-squared (p-value)	0.0004	< 0.05	Significant
2	Age group DI ratio	0.5255	< 0.80	FAIL
3	Age chi-squared (p-value)	0.0001	< 0.05	Significant
4	Top rejection reason	algorithm_risk_score	-	81% of rejections
5	Algorithm rejection vs gender	p = 0.0044	< 0.05	Significant
6	Zip code as proxy	gender p=0.0000, approval p=0.7558	Both < 0.05	Partial — gender link strong, approval link weak

Remediation Recommendations:

- Audit the algorithm risk score model
- Remove or re-encode zip code
- Introduce fairness constraints during model training
- Human-in-the-loop review for borderline cases
- Monitor DI ratios over time

Privacy & Governance Assessment: Privacy Risks

Sensitive Personal Data Detected

- Full name, email, SSN, IP address
- Date of birth and ZIP code
- Detailed spending behaviour data

Privacy Risks

- Multiple direct and indirect identifiers
- Spending behaviour represents sensitive behavioural profiling
- Combination of ZIP code + date of birth + gender increases re-identification risk

Key Issue

- Sensitive PII was originally stored without sufficient protection

```
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1  app_037  f7e04e392a2676c1008fc9dc21750b3176c57c5eed8809...
2  app_215  3263f19e841649667a994a137d9d2444feae15e1b50417...
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```

GDPR & Governance

GDPR Compliance Risks

- No consent tracking mechanism
- Missing data retention policy
- Potential violation of data minimization principle

EU AI Act Implications

- Credit scoring systems classified as High-Risk AI Systems
- Governance Gaps
- No audit trail for automated decisions
- Limited human oversight

Solutions & Recommendations

- Pseudonymization strategy to protect identifiers while preserving data utility
- Data minimization & security measures (GDPR Art. 5 & Art. 32)
- Enable Right to Erasure compliance (Art. 17)

Key Recommendation: Implement privacy-by-design and stronger AI governance controls

Conclusions

Data Quality Controls:
Automated validation to prevent duplicates, invalid entries, and inconsistent values.

Fairness Monitoring:
Regular bias audits using metrics such as the Disparate Impact Ratio.

Privacy & AI Governance:
Pseudonymization, data minimization, audit trails, and human oversight for credit decisions.

Q&A